Robert Ireland

PhD Student · University of Nebraska, Lincoln

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Education

University of Nebraska, Lincoln — *Lincoln, Nebraska*
PhD Student in Mathematics — *2022 - present*
- Advisor: Eloísa Grifo

University of Nebraska, Lincoln — *Lincoln, Nebraska*
M.S. in Mathematics — *2022-2023*

St. Olaf College — *Northfield, Minnesota*
B.A. in Mathematics — *2018-2022*

Research Interests

I am interested in positive characteristic commutative algebra, exploring F-invariants for various classes of rings. Most recently, I have been interested in the F-threshold and Frobenius complexity, particularly for Veronese and determinantal rings.

Publications

Published

Grant B., **Ireland, R**, Koenig E., Wood A. (2023). "Structural properties of group power digraphs". *Involve, a Journal of Mathematics*, Vol. 16, No. 5, pp. 809-832.

Teaching Experience

Spring 2026	**MATH 104 - Applied Calculus**, Instructor of Record
Spring 2026	**MATH 310 - Introduction to Modern Algebra**, Grader
Fall 2025	**MATH 101P - College Algebra Plus Foundations**, Instructor of Record
Fall 2024-Spring 2025	**MATH 101 - College Algebra**, Associate Convener/Instructor of Record
Summer 2024	**MATH 106 - Calculus I**, Instructor of Record
Spring 2024	**MATH 100A - Intermediate Algebra**, Instructor of Record
Fall 2023	**MATH 101 - College Algebra**, Instructor of Record
Spring 2023	**MATH 107 - Calculus II**, Teaching Assistant
Fall 2022	**MATH 107 - Calculus II**, Teaching Assistant

Talks Given

Summer 2026. *A Short Introduction to Frobenius Complexity*. Commutative Algebra Reading Seminar, University of Nebraska-Lincoln.

Spring 2024. *Lower Bounds on Betti Numbers*. Commutative Algebra Reading Seminar, University of Nebraska-Lincoln.

Service & Outreach

Spring 2026 **Commutative Algebra and Adjacent Book Seminar**, Co-organizer — *University of Nebraska, Lincoln*

- Co-organized a weekly reading seminar, for graduate students, committed to learning about integral closure and tight closure theory. Meetings involved discussion of selected readings and problem solving sessions.

Fall 2026 **Math Day Committee**, Member — *University of Nebraska, Lincoln*

- Helped organize ”Math Day,” an event where high school students from across Nebraska come to UNL to take the PROBE exam (Problems Requiring Original and Brilliant Effort) followed by a quiz bowl. In particular, I was responsible for training quiz bowl volunteers on how to perform their roles, helping facilitate the event.

Fall 2024 - Spring 2026 **Graduate Student Advisory Board**, Member — *University of Nebraska, Lincoln*

- Worked with other graduate student advisory board members to act as a liaison between graduate students and faculty, voicing departmental concerns and feedback. Additionally, organized department social events for graduate students and faculty.

Fall 2024 - Spring 2025 **MATH 101 College Algebra**, Associate Convener — *University of Nebraska, Lincoln*

- As associate convener for MATH 101, I worked with first-time instructors to address issues and concerns they may have had with their class. Additionally, I wrote assessments used in class and helped develop the structure of the course for the year.

Fall 2024 - Spring 2025 **Commutative Algebra Reading Seminar**, Co-organizer — *University of Nebraska, Lincoln*

- Co-organized a weekly reading seminar, for graduate students, with a focus on commutative algebra and related fields.

Fall 2023 - Spring 2024 **Graduate Student Seminar**, Co-organizer — *University of Nebraska, Lincoln*

- Co-organized a weekly seminar where graduate students give presentations on mathematical topics, across various subfields, for their fellow graduate students.

Fall 2022 & Fall 2023	**Great Plains Alliance**, Supporting student	*University of Nebraska, Lincoln*

- Accompanied fellow graduate students to smaller universities near Lincoln. Students give mathematics talks for undergraduates and answer questions about graduate student life.

Conferences/Workshops Attended

Oct 2026	**KUMUNU 2026 Commutative Algebra**, Lawrence KS	*University of Kansas*
Aug 2026	**Upcoming Researchers in Commutative Algebra**, Lincoln NE	*University of Nebraska*
May 2026	**PASCA 2.0**, Guanajuato City, MX	*CIMAT*
Mar 2026	**2026 CA+ Conference**, Ames IA	*Iowa State University*
Oct 2025	**KUMUNU 2025 Commutative Algebra**, Columbia MO	*University of Missouri*
June 2025	**SMS 2025: An Introduction to Recent Trends in Commutative Algebra**, Toronto ON	*Fields Institute*
Apr 2025	**RayFest**, Lincoln NE	*University of Nebraska*
Mar 2025	**Commutative Algebra in Mixed Characteristic**, Lincoln NE	*University of Nebraska*
Oct 2024	**KUMUNU 2024 Commutative Algebra**, Columbia MO	*University of Missouri*
Apr 2024	**Recent Developments in Commutative Algebra**, Berkeley CA	*SLMath*